
AUDITING SHORTCUT LEARNING IN SCIENTIFIC MACHINE LEARNING: A THREE-AXIS LEAKAGE DETECTOR VALIDATED AGAINST GROUND TRUTH

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ABSTRACT

Machine learning for the sciences increasingly reports strong predictive performance that, on scrutiny, reflects data leakage or geometric/confound shortcuts rather than the intended physical signal. We introduce **UPAF**, a task-agnostic audit that separates genuine, generalizable signal from three distinct failure modes under a single confidence-interval-based verdict: (i) *memorization leakage*, exposed by a random-versus-group-disjoint split gap; (ii) *target/feature leakage*, exposed by a target-shuffle placebo retrained from scratch; and (iii) *confound shortcuts*, exposed by a confound-only baseline together with a partial-correlation control. Crucially, we validate the auditor *itself* against ground truth: across synthetic stress tests with known leakage knobs and a real fold-switch case with an established null, UPAF recovers the correct verdict on all five reference cases. In particular, it correctly classifies a learned Jacobian stability margin—which on naive evaluation appears to predict protein fold-switching—as a structural (RMSD) shortcut carrying no signal beyond geometry, rather than a genuine thermodynamic predictor. We release UPAF as a reusable instrument and report its operating limits honestly, including a model-flexibility regime in which clean data can be mislabeled as leaking.

Keywords data leakage · shortcut learning · scientific machine learning · model auditing · confound control · partial correlation · reproducibility

1 Introduction

Machine learning is now a standard instrument across the quantitative sciences, and with it has come a recurring failure: reported predictive performance that, on closer inspection, reflects data leakage or spurious shortcuts rather than the scientific signal of interest. This is not a rare accident. A systematic survey found leakage affecting hundreds of papers across seventeen fields, in several cases overturning headline conclusions once corrected [Kapoor & Narayanan, 2023]. The community’s response has largely taken two forms: documentation checklists that ask authors to attest to good practice [Kapoor & Narayanan, 2023], and the adoption of *group-disjoint* evaluation splits—homology-disjoint splits for proteins, scaffold splits for molecules, subject- or site-disjoint splits for clinical and sensor data—so that no group appears in both training and test.

Group-disjoint splitting is a genuine improvement, and it is the single check most practitioners now reach for. But it has a structural blind spot that is rarely stated. A disjoint split detects *memorization* leakage: if a model wins by

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recognizing groups, mixing groups across the split inflates accuracy and the disjoint score collapses. It does *not* detect a *confound shortcut*. A nuisance variable that is correlated with the label—a geometric proxy, a scanner identity, a demographic covariate—generalizes across the split just as well as genuine signal does. The random-versus-disjoint gap stays near zero, the split test reports “clean,” and the apparent signal is nonetheless entirely the confound.

We encountered this concretely. A learned Jacobian stability margin appears, on naive evaluation, to distinguish metamorphic (fold-switching) proteins from rigid ones.² Its performance drop under a homology-disjoint split is small and statistically insignificant—the split test sees nothing wrong. Yet the margin is *beaten* by a baseline that sees only the pairwise structural divergence (RMSD), and retains no explanatory power once RMSD is partialled out: it is a geometric shortcut, not a thermodynamic signal. A split-gap audit—the standard defense—would have passed it. What was missing was a test aimed at confounds rather than memorization.

This motivates two requirements. First, an audit must separate the failure modes it is asked to rule out: memorization leakage, target/feature leakage, and confound shortcuts are distinct, and the most common check is blind to one of them. Second, the audit must itself be trustworthy. An auditor that has never been checked against cases with known ground truth is just another unvalidated model; its verdicts cannot be taken on faith—least of all when they are convenient.

Contributions.

- **A three-axis audit (UPAF)** that distinguishes memorization leakage (a random-vs-group-disjoint split gap), feature/target leakage (a from-scratch target-shuffle placebo), and confound shortcuts (a confound-only baseline with a partial-correlation control), combined into a single confidence-interval-based verdict.
- **The confound blind spot, made explicit and measured:** we show that the widely used split-gap test cannot detect confound shortcuts, and that only the confound axis recovers the correct verdict on a real case that the split test passes.
- **A ground-truth self-validation gate:** before any downstream claim, the auditor must reproduce the known verdict on five reference cases—synthetic stress tests with controlled leakage knobs and a real null; it passes all five.
- **Calibration and honest operating limits:** an independent-knob synthetic benchmark establishing sensitivity and specificity, real-data case studies spanning all three verdicts, and a documented regime in which clean data can be mislabeled.

2 Related Work

Leakage and reproducibility in ML-based science. The pitfalls of leakage have a long history [Kaufman et al., 2012], and their prevalence in scientific ML was documented at scale by Kapoor & Narayanan [2023], who catalog eight leakage types and propose model info sheets—structured questionnaires authors complete to attest that each type has been considered. Our aim is complementary but distinct: rather than a documentation standard applied by the author, we provide a *runnable instrument* that takes a model and dataset and returns a verdict, and that localizes *which* failure mode is present. Where a checklist asks whether a confound was considered, our audit measures whether the model’s signal survives controlling for it.

Shortcut learning. That networks exploit superficial cues that fail to generalize is well established as a phenomenon [Geirhos et al., 2020]. We contribute on the detection side: a task-agnostic procedure that separates a shortcut driven by a measurable confound from genuine generalizable signal.

Evaluation splits and statistical controls. Group-disjoint splits and permutation/placebo tests are standard individual tools, as is partial correlation for removing a nuisance covariate. Our contribution is not any single test but their integration into a coherent verdict, the explicit demonstration that the split-gap test is blind to confounds, and—unusually—the validation of the combined procedure against ground truth.

Automated leakage detection. Recent automated approaches target different objects: static analysis of ML *code* for leakage-inducing patterns [Alturayef & Hassine, 2025], and detection of benchmark *contamination* in large language models via n-gram or perplexity signals [Xu et al., 2024]. Neither addresses task-level confound shortcuts in a trained scientific predictor, which is our focus.

²The margin audited here is the stability readout of a companion volume-preserving deep-equilibrium model [Companion S-DEQ, 2026]; we cite it as the source of the audited quantity. *Optional cross-citation—remove if the companion preprint is not posted.*

3 The UPAF Audit: A Three-Axis Leakage Detector

Reported predictive performance in scientific ML conflates at least three distinct phenomena: genuine generalizable signal, leakage from train–test dependence, and reliance on a confounding nuisance variable that correlates with the label but is not the intended target of study. A single scalar (AUROC, R^2) cannot disentangle these. UPAF instead computes a small, fixed battery of quantities and applies a pre-registered decision rule whose branches correspond to the three failure modes.

3.1 Measured quantities

Given a model class, a dataset with grouping variable g (e.g. sequence family), and an optional set of named confound features $\mathcal{C}$, UPAF estimates the following, each over $S \geq 5$ seeds with a 1000-sample bootstrap 95% confidence interval:

- B — a *trivial baseline* (majority class / mean predictor);
- C — a *confound-only baseline*, trained using *only* the features in $\mathcal{C}$ (e.g. RMSD, length, composition);
- D — performance under a *group-disjoint* split (no group appears in both train and test);
- R — performance under a *random* (group-mixing) split;
- P — a *target-shuffle placebo*: training labels/targets are permuted, the model is retrained *from scratch*, and evaluated on the true disjoint test set;
- PC — the partial-correlation p -value between the disjoint-split predictions and the labels, controlling for $\mathcal{C}$, aggregated across seeds by the median.

All exceedance comparisons (e.g. “ $D > B$ ”, “ $R > D$ ”) are decided by confidence-interval non-overlap combined with a minimal effect size (0.05 for classification, 0.15 for regression), rather than by a hard point-estimate threshold; this choice is examined in Section 5.1.

3.2 The three axes

Each axis isolates one failure mode:

Axis 1 — Memorization (split gap $R - D$). If a model exploits within-group structure, mixing groups across train and test inflates R while the disjoint split D collapses. A significant gap $R \gg D$ therefore signals memorization leakage. Confounds, by contrast, are present under *both* splits and do not produce this gap.

Axis 2 — Target/feature leakage (placebo P). If predictive power survives permuting the training targets, the model is reading the label through a channel that bypasses supervision (a direct encoding of the target in the inputs). A placebo that remains well above baseline signals such leakage.

Axis 3 — Confound shortcut (C and PC). A model may generalize across disjoint groups yet still be a shortcut: its signal may be fully explained by a nuisance confound. We detect this by checking whether the model significantly exceeds the confound-only baseline ($D > C$) and whether its predictions retain explanatory power after partialling out the confound ($PC < 0.05$). This axis is essential and is the one most often omitted: the split gap (Axis 1) is *blind* to confound shortcuts, because a confound generalizes across splits and yields $R \approx D$.

3.3 Unified verdict

Given $\{B, C, D, R, P, PC\}$ with their CIs, UPAF returns one of NULL, LEAKAGE, SHORTCUT, SIGNAL-GENUINE, or UNDERPOWERED by the following pre-registered rule:

1. **No generalizable signal** ($D \approx B$): return LEAKAGE if $R \gg D$ significantly, else NULL.
2. **Disjoint signal present** ($D > B$):
 - (a) if $D \not> C$ or $PC > 0.05 \Rightarrow$ SHORTCUT (signal explained by the confound);
 - (b) else if $R \gg D$ or P remains high $\Rightarrow$ LEAKAGE;
 - (c) else $\Rightarrow$ SIGNAL-GENUINE.
3. otherwise $\Rightarrow$ UNDERPOWERED.

We stress that the verdict is a *hypothesis about the model*, never a target to be reached; the auditor never modifies the auditee to obtain a desired label.

4 Validating the Auditor: A Ground-Truth Self-Test

An audit is only useful if it returns the correct verdict on cases whose ground truth is independently known. Before applying UPAF to any task of unknown status, we require it to pass a fixed battery of five reference cases. If it fails any, the tool is reported as unvalidated rather than tuned to pass.

Table 1: Ground-truth self-validation. UPAF is required to match the known verdict on five reference cases before any downstream claim. All five pass. Synthetic knobs α, β, γ are defined in Section 5; the fold-switch margin’s null status is established independently.

Reference case	Required	UPAF	Mechanism
Fold-switch margin (real)	SHORTCUT/NULL	SHORTCUT	$D=0.567 < C=0.876$; PC $p=0.93$
Synthetic $\alpha=1$ (feature leak)	LEAKAGE	LEAKAGE	$P=1.00 \gg B=0.50$
Synthetic $\beta=1$ (memorization)	LEAKAGE	LEAKAGE	$R=0.99 \gg D=0.64$
Synthetic $\gamma \geq 0.8$ (confound)	SHORTCUT	SHORTCUT	$C=1.00=D$; PC $p \rightarrow 1$
Synthetic $\alpha=\beta=0$ (clean)	clean	SIGNAL-GENUINE	no significant gap; P at chance

The fold-switch case is the decisive real-data check. A naive evaluation of a learned Jacobian stability margin appears to separate metamorphic from rigid proteins, but the margin is significantly *beaten* by an RMSD-only baseline on disjoint folds ($D=0.567$ vs. $C=0.876$) and retains no explanatory power once RMSD is partialled out (PC $p=0.93$). UPAF therefore returns SHORTCUT, consistent with the established result that the margin is a geometric proxy rather than a thermodynamic signal. Notably, an audit using the split gap alone (Axis 1) would *mislabel* this case as genuine, since the RMSD shortcut generalizes across folds ($R \approx D$); only the confound axis (Axis 3) recovers the correct verdict. This case directly motivates the three-axis design.

5 Synthetic Calibration

To characterize each axis quantitatively, we generate synthetic data with three independent leakage knobs and verify that the relevant detection statistic responds monotonically, while remaining clean when the knob is off.

Knob α (feature leak). An input feature encodes the label with strength α . The placebo gap $P-B$ rises monotonically: -0.01 ($\alpha=0$, clean) $\rightarrow 0.05 \rightarrow 0.30 \rightarrow 0.47 \rightarrow 0.50$ ($\alpha=1$), with LEAKAGE declared for all $\alpha \geq 0.2$.

Knob β (group memorization). Within-group label agreement rises with β . The split gap $R-D$ rises 0.02 ($\beta=0$) $\rightarrow 0.04 \rightarrow 0.06 \rightarrow 0.15 \rightarrow 0.35$ ($\beta=1$), crossing into LEAKAGE at $\beta \approx 0.6$ under the CI-plus-effect-size rule.

Knob γ (confound shortcut). A confound predicts the label with strength γ while the genuine feature is absent. The confound-only baseline C rises $0.50 \rightarrow 0.80 \rightarrow 1.00$ and, at $\gamma \geq 0.8$, fully matches D with non-significant partial correlation, yielding SHORTCUT.

Honest behavior of the γ knob. The γ sweep is not cleanly monotone into SHORTCUT: at intermediate γ the injected confound also induces split sensitivity, so the verdict passes through LEAKAGE ($0.2 \leq \gamma \leq 0.6$) before reaching SHORTCUT at the endpoints. The endpoints—genuine at $\gamma=0$, shortcut at $\gamma=1$ —are correct, but we report this intermediate mixing rather than presenting an idealized curve.

5.1 When the auditor fails: a model-flexibility false positive

On fully clean data ($\alpha=\beta=0$), a sufficiently flexible auditee model can memorize group identity even from random labels, inflating R over D and producing a spurious LEAKAGE call. We *do not* remove this by tuning the model; instead we resolve the spurious call principledly, by deciding the split gap via CI overlap and a minimal effect size rather than a raw cutoff, and we characterize the residual risk directly: sweeping decision-tree depth over $\{2, 3, 5, 8, \infty\}$ on clean data, the split gap stays within noise and all configurations are correctly returned as SIGNAL-GENUINE. We note this flexibility-versus-specificity trade-off as an operating limit of the audit.

6 Real-Data Case Studies

We apply UPAF to five real tasks spanning two domains: protein biophysics (fold-switch stability, liquid–liquid phase separation, signal-peptide detection, subcellular localization [Thummuluri et al., 2022, Teufel et al., 2022]) and clinical tabular prediction (heart disease across four hospital sites [Detrano et al., 1989]). The five tasks were pre-specified and audited blind under a single-open lock; we report the verdict each received, not a tuned outcome. All three verdict types appear, and the baseline B is fixed at its theoretical chance value (0.5 for classification) rather than estimated.

Table 2: Blind audit of five real tasks (5 seeds, example-level bootstrap 95% CI; B fixed at chance). C : confound-only baseline; D : group-disjoint; R : random split; P : target-shuffle placebo; PC: partial-correlation p controlling for the confound. Confounds are RMSD (A), sequence length and amino-acid composition (B,D,E), and demographic/site covariates (F).

Task	Domain	C	D	R	P	PC p	Verdict
A: fold-switch	protein	0.876	0.567	0.589	0.464	0.93	SHORTCUT
B: LLPS	protein	0.507	0.591	0.672	0.519	$< 10^{-12}$	LEAKAGE
F: heart disease	clinical	0.691	0.774	0.883	0.364	$< 10^{-42}$	LEAKAGE
D: signal peptide	protein	0.868	0.990	0.994	0.478	$< 10^{-300}$	SIGNAL
E: localization	protein	0.687	0.837	0.848	0.493	$< 10^{-300}$	SIGNAL

6.1 A confound shortcut that the split test misses

Task A is the case that motivates the entire three-axis design. A learned Jacobian stability margin appears to separate metamorphic from rigid proteins, and it shows *no significant memorization gap*: the random-split and disjoint scores ($R=0.589$, $D=0.567$) differ by only 0.02, with overlapping confidence intervals and a gap below the detection threshold. By the split-gap test—the check most practitioners rely on—the margin looks clean. Yet a baseline that sees only the pairwise RMSD beats it outright ($C=0.876 \gg D=0.567$), and once RMSD is partialled out the margin retains no explanatory power (PC $p=0.93$). The confound axis returns SHORTCUT: the margin is a geometric proxy, not a thermodynamic signal. A split-only audit would have certified it; only Axis 3 recovers the truth. This is the empirical core of the blind spot identified in Section 1.

6.2 Memorization leakage across two domains

Tasks B and F are caught by the memorization axis, in two different domains. In LLPS classification (B), the random split inflates performance ($R=0.672$) over the disjoint estimate ($D=0.591$), a gap that exceeds the effect-size threshold and yields LEAKAGE from sequence-similarity grouping. In clinical heart disease (F), the same pattern appears across hospital sites: a random split reaches $R=0.883$, but holding out an entire site drops the honest estimate to $D=0.774$. Notably, F still carries genuine signal beyond its confound ($D > C$, PC significant); the verdict is LEAKAGE because the *evaluation*—not the model—is contaminated: a random split over grouped clinical data over-reports by roughly 0.11 AUROC. The audit thus flags an over-optimistic protocol that is widespread in tabular scientific ML, recovering the same failure mode the auditor was built to catch in proteins.

6.3 Genuine signal: specificity on real data

An auditor that labels everything as leaking is useless. Tasks D and E are clean positive controls. Signal-peptide detection and cytoplasm-versus-nucleus localization both exceed their composition-and-length confound by a paired margin whose bootstrap interval excludes zero ($D-C$: $[0.106, 0.137]$ and $[0.140, 0.159]$), retain strong partial correlation after controlling for it, show negligible split gaps ($R-D \leq 0.011$), and collapse to chance under the target-shuffle placebo. Both receive SIGNAL-GENUINE. Their presence establishes specificity: on real biological data with genuine generalizable structure, UPAF does not raise a false alarm.

Summary. Across five real tasks and two domains, UPAF returns all three non-trivial verdicts—a confound shortcut invisible to the split test (A), memorization leakage in both protein and clinical data (B, F), and genuine signal (D, E)—without any task being tuned toward its verdict.

7 Limitations

Real-data evidence for the confound shortcut is limited. The central claim—that the split-gap test is blind to confound shortcuts—is demonstrated on one real task (the fold-switch margin, Section 6.1) together with a controlled synthetic confound knob (Section 5). The synthetic sweep shows the blind spot is not an artifact of a single dataset, but we do not yet have a second real-world confound shortcut in a different domain; how common this failure mode is across scientific ML is left to future work.

Verdicts characterize a (model, hyperparameter) pair, not a task in the abstract. On one regression task we observed that the verdict depended on the auditee’s regularization strength: strong regularization yielded a clean verdict while weak regularization produced a leakage verdict, because the random-vs-disjoint gap grows once the model is allowed to fit group-specific structure. We therefore fix auditee hyperparameters by nested cross-validation, selected only on training folds, and read each verdict as a statement about a specified model and training procedure rather than about the task in isolation. An audit should report the auditee configuration alongside its verdict.

Decision thresholds are design choices. Verdicts use pre-registered effect-size thresholds (0.05 classification, 0.15 regression) combined with confidence-interval non-overlap. These are deliberately conservative, favoring low false-positive rates, so weak leakage may be under-flagged. A threshold-sensitivity analysis shows most verdicts are stable across a wide band—genuine signal and the confound shortcut are threshold-independent—while a few cases sit nearer the boundary; one leakage verdict, for instance, has a split gap only moderately above the threshold. We report this sensitivity rather than relying on a single cutoff, and we never re-select the threshold to obtain a preferred verdict.

The synthetic confound knob is not cleanly monotone. In our controlled generator, increasing the confound strength γ does not move the verdict directly from genuine to shortcut: at intermediate strengths the injected confound also induces split sensitivity, so the verdict passes through leakage before reaching shortcut at the endpoints. The endpoints are correct; we report the intermediate mixing rather than presenting an idealized curve.

Auditor flexibility and false positives on clean data. A sufficiently flexible auditee can memorize group identity even from randomized labels, inflating the split gap on clean data. We mitigate this with the interval-plus-effect-size rule and characterize the residual risk with a capacity sweep (Section 5), but the trade-off between auditee flexibility and specificity is intrinsic and should be kept in mind when configuring an audit.

Scope. Real-data validation spans protein biophysics and clinical tabular prediction. Although the audit is task-agnostic by construction, its behavior on other modalities (images, time series, text) is untested and left to future work.

8 Conclusion

Reported performance in scientific machine learning routinely conflates three different things: genuine generalizable signal, leakage from train–test dependence, and reliance on a confounding nuisance variable. We presented UPAF, a task-agnostic audit that separates these three under a single confidence-interval-based verdict, and—unusually for an evaluation tool—we validated the auditor itself against ground truth, requiring it to recover the known verdict on five reference cases before trusting it downstream.

Our central finding is a blind spot in standard practice: the group-disjoint split, the leakage check most practitioners rely on, cannot detect a confound shortcut, because a confound generalizes across the split just as genuine signal does. We make this concrete on a real case—a learned protein stability margin whose split-test performance drop is statistically insignificant, yet which is fully explained by a geometric confound—and characterize it across strengths with a controlled synthetic generator. Only an explicit confound axis recovers the correct verdict.

Applied blind to five real tasks across protein biophysics and clinical tabular prediction, UPAF returns all three non-trivial verdicts without any task being tuned toward its outcome, and on clean data it does not raise false alarms. We report its limits honestly: a single real confound-shortcut case, sensitivity of borderline verdicts to the auditee’s configuration and to threshold choices, and a synthetic confound knob that is not cleanly monotone.

We release UPAF as a reusable instrument, with a single-command reproduction of every reported number. More broadly, we argue that an auditor which has never been checked against ground truth should not be trusted—and that “which failure mode is this?” is a more useful question for scientific ML than “how high is the score?”

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A Reproducibility and Integrity Protocol

We designed the audit and its evaluation to be reproducible and resistant to post-hoc adjustment.

Single-open configuration lock. Each audited task is registered once under a configuration hash computed over the auditee model class and hyperparameters, the feature matrix, the labels, *and* the group assignments used for the disjoint split. Any subsequent change to these—including how groups are formed—produces a hash mismatch logged as a reset, preventing silent alteration of the split, model, or data after a verdict is observed. Effect-size thresholds (0.05 classification, 0.15 regression) were fixed before the audits and are not re-selected per task.

Deterministic splits. An early version of the fold-switch group-disjoint split derived sequence families by iterating over a hash-seeded set, whose order depends on the process hash seed, making the disjoint score irreproducible across runs. We made all group construction deterministic (sorted ordering) and verified this directly: running the complete pipeline under two different process hash seeds yields byte-identical metrics, confidence intervals, and verdicts. We report the deterministic values, including where they are less favorable than an earlier seed-dependent draw—the fold-switch disjoint AUROC is 0.567, not a more separated value obtained under one seed; the SHORTCUT verdict is unchanged across all seeds.

An excluded task. One regression task is excluded from our results: its regularization strength was changed after its audit metric had been observed, which our protocol treats as a broken blind regardless of the resulting verdict. We report the exclusion rather than the numbers. The calibration sweeps register a reset at each sweep point because the synthetic dataset changes per point; these are artifacts of routing a parameter sweep through a single-task lock, not post-hoc reconfigurations, and are logged transparently.

Release. We release an open-source package that reproduces every reported number with a single command, with data provenance for all real datasets (public homology- and site-disjoint partitions). Wall-clock durations and access timestamps are the only fields that vary between runs.